Heuristic-Driven Routing: A Technical and Social Analysis of the A* Algorithm

Francis Kyle Sunga

School of Computing, Binghamton University

CS 301: Ethical, Social, and Global Issues in Computing

Dr. George Weinschenk

October 10, 2025

Abstract

This paper evaluates the key technical behaviors and implications of the A* pathfinding algorithm in comparison to Dijkstra's algorithm. Dijkstra's pathfinding method is the basis for most pathfinding algorithms, guaranteeing the shortest path, yet it is unable to adapt to real-world scenarios. A* improves upon Dijkstra by combining the cost traveled with calculated estimates that predict the remaining distance; this allows A* to respond more effectively to environmental factors such as traffic, emergency responders, and convoy routes. A* thrives in computational situations where delayed rerouting can influence the safety and survival of first responders, refugees, and patients. By examining cases in which traditional algorithms fall short, this paper highlights that heuristic-driven approaches better support navigation systems that must operate quickly, reliably, and ethically in high-risk conditions. The results make it clear that A* excels beyond just efficiency, reinforcing that modern navigation systems depend on real-time information to maintain reliability and safety.

Heuristic-Driven Routing: A Technical and Social Analysis of the A* Algorithm

In today's world, the use of navigation systems has grown exponentially; every time a user opens Google Maps or Waze, an algorithm works to find the most optimal route. Tiny decisions, such as which turn to take and how fast to travel, dictate the wear on certain roads, fuel use, and city accessibility. The constant trust in these algorithms raises questions about fairness and control within modern computing. Dijkstra and A* both find the fastest path between two points, but A* uses extra data to predict distance ahead of time, which allows computers to compute routes more quickly and efficiently. Compared to Dijkstra's Algorithm, the A search algorithm provides better computational efficiency and scalability, resulting in more successful—but potentially biased—navigation systems.

Technical Background for Alternative Algorithm

Dijkstra's algorithm sets the foundation for most pathfinding and route systems that apps like Google Maps use today. Dijkstra's algorithm works by choosing a starting point and checking every possible route; Dijkstra then picks the route with the shortest distance. Every connection between nodes carries a value called its weight. Each connection holds key values such as the route's distance, time, and cost factors. The algorithm visits every node one at a time, updating the total cost until the algorithm reaches the user's destination. This method finds the most accurate route, but slows down as the network expands, limiting its use in larger systems.

When used in real navigation, Dijkstra's process treats each route as fixed. Transportation systems researcher Vladimir Milin of the university of Split modeled this behavior in maritime navigation between ports, where each edge weight reflects the distance between two points

(Milin et al. 2025). Their study proves that Dijkstra's algorithm searches for the shortest path, ignoring factors such as wind direction or current speed. The researchers represented this simple logic with the base distance formula shown below:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \tag{1}$$

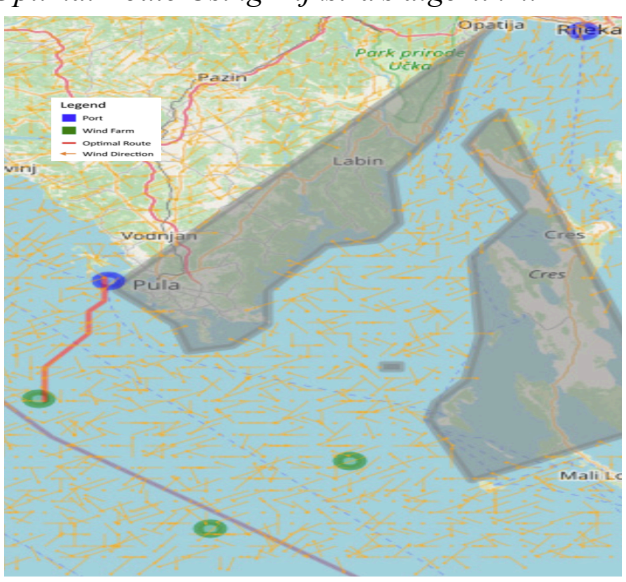
This equation measures the shortest possible distance between nodes. However, the study also introduced an improvised version that accounted for environmental factors that Dijkstra did not.

$$dist \times (1 + \frac{resistance \times windspeed}{20}) \tag{2}$$

Here, resistance measures how much environmental conditions like wind or current affect travel efficiency, adjusting the travel cost accordingly (Milin et al. 2025). Dijkstra's method ignores this adjustment and relies on the base distance term, which means Dijkstra assumes ideal conditions. As a result, Dijkstra produces accurate routes in theory, but fails when those conditions shift in real time.

Figure 1

Optimal Route Using Dijkstra's algorithm.



Although Dijkstra's algorithm performs well in simple environments, the algorithm struggles with scalability and adaptability. Computer scientist Dr. Emilio Ferrara of the Univarsity of Southern California notes that shortest-path routing overloads certain roads, leading to uneven traffic patterns (Ferrara et al., 2025). This emphasizes the need for algorithms like A* that incorporate predictable data, reducing computation time and adapting to real-world conditions.

Support

Technical Description of Primary Algorithm

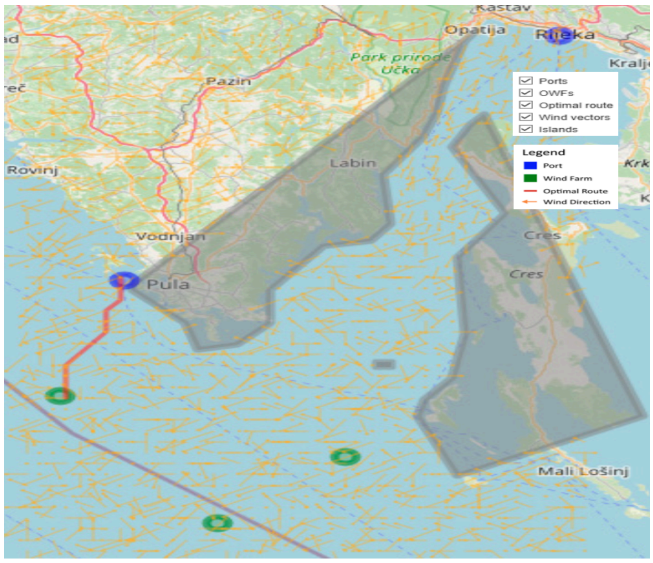
A* builds on Dijkstra's approach, but adds a way for the search to move directly toward the destination. Instead of treating each node the same, A* estimates how far each node is from the destination and uses that information to guide its direction. Like Dijkstra's algorithm, A* contains a weight, but included in that, it also holds a predicted cost called a heuristic. As the algorithm moves through the network, it updates both the distance traveled and the distance remaining. A* combines these two values by using the function shown in (3):

$$f(n) = g(n) + h(n) \tag{3}$$

Equation (3) captures how A* travels to its final destination. The term g(n) refers to the cost spent to get to said node, while h(n) represents the prediction cost to get to the final destination. Together, they allow A* to focus on making efficient decisions, giving it a clear advantage in larger, more dynamic systems. Stanford robotics professor Marco Pavone, a leading researcher in autonomous systems and transportation optimization, emphasizes that predictive routing greatly improves efficiency in large-scale networks (Jalora et al., 2023)

Figure 2

Optimal Route Using A Algorithm*



This advantage appears clearly when applied to a real-world scenario. Figure 2 demonstrates this by showing how A* takes into consideration island obstacles, port locations, and wind vectors. By taking into account these environmental factors directly into its cost and heuristic functions, A* prioritises routes that minimize distance and environmental factors. A clear example is wind resistance. Equation (3) incorporates wind resistance directly into g(n) and h(n), meaning that A* actively avoids routes where stronger wind resistance can increase fuel intake and travel time. A* also considers port proximity, accounting for how far ports are from the nodes, producing a more efficient docking path when approaching coastal regions. The algorithm takes into account how far ports are from the nodes, favoring a safer and more efficient docking path when approaching coastal regions.

Figure 3

Optimal Route With Wind Direction

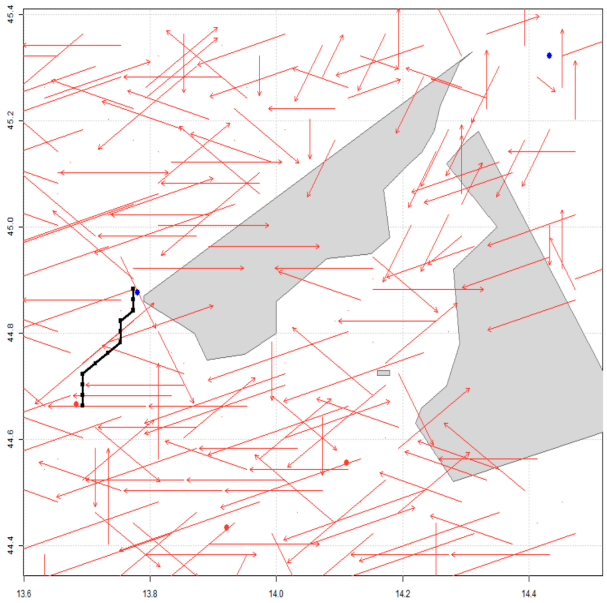


Figure 3, displayed above, highlights how A* behaves and adapts to real forces rather than grid conditions. Each red arrow displayed represents a local wind vector, indicating both the direction and magnitude of the wind vector. The route actively avoids areas where the arrows point directly against the vessel's heading, which reduces drag and travel time. A* shifts the vessel toward segments where wind direction would positively affect the vessel or cause minimal cross-current interference. In addition, the figure also accounts for spatial awareness near landmasses. The figures displayed in gray act as land, and the A* path intentionally navigates along the route while keeping a safe distance from land. By blending geometry with real-time factors, A*'s path makes sense both mathematically and practically. The behaviors in Figure 3

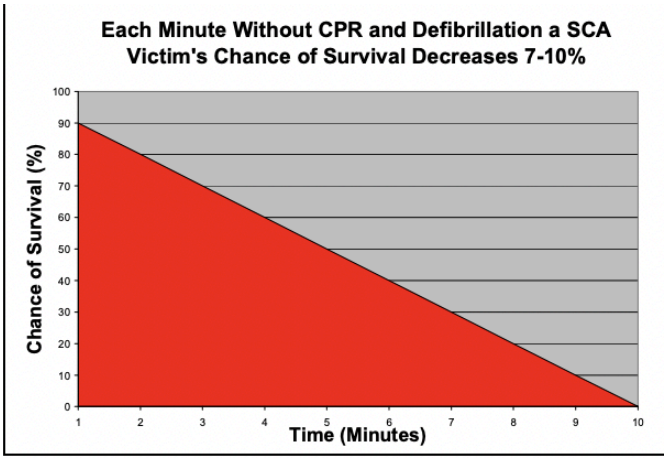
illustrate this clearly. Milin et al. (2025) emphasize that these adjustments in the algorithm's path are not arbitrary; they are directly derived from how the algorithm recalculates g(n) and h(n) at every node. As the environment shifts, the cost model recalculates in real time. This makes the search inherently adaptive; instead of committing to a predetermined path, A* puts safety and efficiency at the forefront, continually responding to updates of cost conditions and other earthly factors. These measures underline A*'s strengths as a navigation system, an algorithm capable of integrating physical constraints directly into its cost evaluations to generate a reliable and operationally realistic path.

Social Impact

In today's modern economy, routing and navigation don't just serve individual people; they influence entire industries. Consider Google Maps or Uber, apps that direct millions of users daily. According to a study by the National Institute of Standards and Technology, location-based navigation generated $218 billion in the U.S. between 2007 and 2017. For instance, Uber and Lyft depend on real-time pathfinding to assign drivers, routes, and fare costs across millions of users. These apps rely on algorithms such as A* to take into account traffic congestion, fuel efficiency, and potential risks. Beyond just these apps, A* integrates directly into public safety and emergency response systems, where emergency services can dispatch the nearest responders while accounting for traffic conditions and road closures. This capability in itself directly saves many lives.

Table 1

Survival Probability Declines 7–10% Per Minute Without CPR and Defibrillation



According to the American Heart Association, a nationally recognized medical research organization, survival rates in cardiac arrest decrease by 7-10% for every minute without defibrillation (Oregon Legislative Assembly, 2013). A* Pathfinding shortens emergency response times, in instances where a few minutes can save a person's life. This statistic highlights the consequences of routing. Emergency vehicles need to operate in environments where seconds matter. Table 1 illustrates this; by the 7th minute without receiving defibrillation, a patient's chance of survival has already fallen to 30%. Algorithms like Dijkstra,which scan and process every node, create unnecessary delays that put people at risk. In contrast, A* reduces that and prioritizes the most promising path, allowing dispatch to arrive on the scene in significantly less time.

The social impacts become clear when these algorithms are tested in large city-wide networks. Large metropolitan areas like New York City receive roughly 4,400 emergency service calls daily, and cutting down even 30-60 seconds off each dispatch can directly relate to hundreds more lives saved annually. When embedded into emergency services, A*'s heuristic search feature significantly helps responders avoid congested intersections, increasing the probability of a patient's life potentially being saved or lost. Beyond user navigation and emergency response, A* also plays a role in humanitarian logistics. In war-torn areas,

organizations such as the United Nations World Food Programme, the world's largest humanitarian logistics agency, reports that humanitarian convoys must constantly reroute due to shifting front-line conditions (World Food Programme USA, 2024). In these environments, traditional routes and roads can be destroyed or dangerous, and navigation methods such as Dijkstra's algorithm frequently fall short because they fail to incorporate hazards that change hour by hour. Now A* evaluates these hazards in real time, taking into account destroyed infrastructure, restricted airways and naval paths, shifting of front lines, and landmines. According to the World Food Programme (2025), humanitarian convoys must constantly reroute due to heavy insecurity from military activities. WPF notes that these conditions require dynamic navigation systems capable of calculating safe paths in real time so that food and medical supplies can reach isolated populations without interruptions ("Gaza Convoy Incident," 2025). A*'s ability to recalculate safe paths allows convoys to avoid blocked or dangerous routes and still reach communities in need. This means that communities cut off by warfare received aid and supplies sooner and more consistently. The impacts of this algorithm go farther than just taking users to work and improving efficiency; it increases the rate of survival for refugees and families who depend on timely aid for basic sustenance.

Conclusion

The comparison between Dijkstra and A* highlights a benchmark in how modern navigations must operate. Dijkstra provides accuracy, compromising speed and struggle in real-world scenarios. A* on the other hand adapts to real-time conditions, maximising efficiency and safety directly into its calculations. Modern systems that utilize A*'s heuristics account for continuous updates, enabling first responders to reach affected areas faster than those relying on algorithms like Dijkstra. The findings throughout this analysis further emphasize that performance depends more than just finding the shortest path. In emergency response networks, city-scale transportation systems, the delay of seconds can shape the outcome of first responders, refugees, and civilians. A*'s approach reduces these delays by prioritizing routes that reflect hazards and geographic constraints.

For computing professionals, the significance of this algorithm goes far past the design. Developers, managers, and engineers must share responsibility for ensuring that the algorithms they build operate safely, ethically, and reliably for when people depend on them in their most vulnerable state. As technologies continue to expand, the risks and patterns that arise must be the main focal point moving forward. This analysis supports the thesis that A* offers both technical and social advantages over traditional approaches, and that navigation systems being developed in the future must continue moving toward adaptive models grounded in safety, transparency, and fairness for all users. How we choose to design these algorithms dictates how communities move, respond, and stay safe.

References

Ferrara, A., García-Soriano, D., & Bonchi, F. (2025). *Beyond shortest paths: Node fairness in route recommendation*. *Proceedings of the VLDB Endowment, 18*(9), 3230–3242. https://doi.org/10.14778/3746405.3746440

Jalota, D., Solovey, K., Tsao, M., Zoepf, S., & Pavone, M. (2023). *Balancing fairness and efficiency in traffic routing via interpolated traffic assignment*. *Autonomous Agents and Multi-Agent Systems, 37*(32). https://doi.org/10.1007/s10458-023-09616-7

Milin, V., Stanivuk, T., Skoko, I., & Bulić, T. (2025). *Dijkstra and A* algorithms for algorithmic optimization of maritime routes and logistics of offshore wind farms*. *Journal of Marine Science and Engineering, 13*(1863). https://doi.org/10.3390/jmse13101863

Oregon Legislative Assembly. (2013). *Automated external defibrillator requirements for public buildings* (Committee Meeting Document 4759). https://olis.oregonlegislature.gov/liz/2013R1/Downloads/CommitteeMeetingDocument/4759

Sattaratnamai, S., Wongsai, N., & Ratanavaraha, V. (2020). *Comparison of shortest path algorithms for route selection in transportation networks*. *International Journal of Intelligent Transportation Systems Research, 18*(3), 442–455. https://www.mendeley.com/catalogue/8152e931-4f00-3448-a138-3fad8c165243/

Slavin, K. (2011). *How algorithms shape our world* [Video]. TED Conferences. https://www.ted.com/talks/kevin_slavin_how_algorithms_shape_our_world

World Food Program USA. (2024). *WFP statement on Gaza convoy incident*. https://wfpusa.org/news/gaza-convoy-incident-wfp-statement/